Ismail Sarfaraz

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SUMMARY

Data Scientist with 2+ years of experience developing and validating ML solutions across healthcare and fintech domains. Strong in Python, SQL, Spark, Tableau and statistical modeling, with hands-on experience in feature engineering, experimentation, and production ML pipelines. Improved downstream model AUC from 0.76 to 0.84 and built pipelines integrating 12+ clinical data sources using Databricks and Snowflake.

PROFESSIONAL EXPERIENCE

George Washington University — **Mar 2025 - May 2026**
Research Analyst — *Washington, DC, USA*

- Diagnosed root-cause failure in budget forecasting models by reconciling $2M+ in historical funding data, restoring projection accuracy.
- Authored analytical SQL queries & Python validation pipelines on multi-source grant & cohort data, powering live executive dashboards.
- Ran ANOVA across demographic and program variables to surface retention drivers in longitudinal studies, closing a 25% attrition gap.

Tata Consultancy Services (TCS) — **Jun 2023 - Jun 2024**
Data Scientist — *Hyderabad, TG, India*

- Built end-to-end ML feature pipelines using PySpark, engineering curated tables to support predictive modeling and advanced analytics.
- Drove downstream model AUC from 0.76 to 0.84 by enriching curated feature tables and resolving quality gaps in upstream clinical data.
- Validated new pipeline via 30-day A/B test, tracking feature drift, model lift & SLA compliance against legacy heuristics before cutover.
- Architected Medallion ETL pipeline (Databricks/Airflow) syncing 12+ PHI feeds to Snowflake, enabling real-time risk-scoring inference.
- Designed HIPAA-compliant validation framework (Great Expectations) on feature tables, cutting quality incidents 20% in production.

Tata Consultancy Services (TCS) — **Jan 2023 - May 2023**
Data Analyst Intern — *Hyderabad, TG, India*

- Conducted Pearson correlation analysis across 20+ metrics to define 15+ strategic KPIs aligned with core business strategy & risk metrics.
- Built Tableau dashboards (LOD/parameters) powering self-serve analytics on market segments & exposure to 12+ business stakeholders.
- Streamlined extraction pipelines (Python/SQLAlchemy) serving 12+ stakeholders; cut query latency by 30% enabling real-time analytics.

PROJECTS

Sentinel-Health: FDA Device Adverse-Event Early-Warning System - ***Link to project***

- Built an FDA adverse-event surveillance pipeline over 4.6M MAUDE reports, resolving 7,037 manufacturer spellings to 4,433 entities.
- Extracted failure modes from clinical narratives, evaluating rules vs LLM on 300 hand-labeled reports (47.3% vs 45.0% accuracy).

AWS CloudGuide - Cloud Architecture Assistant - ***Link to project***

- Architected hybrid RAG using dense (bge-m3) & sparse (BM25) retrieval via RRF, resolving semantic ambiguity across 100k+ AWS pages.
- Scaled batched GPU pipeline (PyTorch, T4) handling 800k+ vectors using recursive chunking for context preservation & retrieval accuracy.
- Deployed LLM-powered conversational Q&A system (Qwen-2.5-7B) via Hugging Face achieving sub-200ms latency for cited answers.

Cricket Live Match Win Predictor - ***Link to project***

- Developed a live, ball-by-ball win probability predictor for IPL matches using an XGBoost model trained on 500+ historical matches.
- Stabilized early-game prediction volatility by designing a confidence-weighted blend-in algorithm, reducing forecast uncertainty by 60%.

EDUCATION

George Washington University — **Aug 2024 - May 2026**
Master's, Data Science — *GPA: 3.95*
Relevant Coursework: Data Mining, Deep Learning, Mathematics in Machine Learning, Data Analytics, Data Visualization, NLP

TECHNICAL SKILLS

Languages: Python (Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow), SQL, R, SAS
Data & Cloud: PySpark, AWS (SageMaker, EC2, S3, Glue, Lambda), Databricks, Snowflake, BigQuery, Airflow, Docker, Tableau, ETL/ELT
ML & Statistics: A/B Testing, Hypothesis Testing, Regression Analysis, ANOVA, Causal Inference, Feature Engineering, Statistical Modeling
GenAI & LLMs: LangChain, Hugging Face, RAG, Vector Databases (Pinecone, Chroma, FAISS), Model Fine-tuning, Prompt Engineering